

Normalisasi linear menggunakan Z-Score Normalization

<i>Nama</i>	<i>Fitur 1</i>	<i>Fitur 2</i>
<i>X1</i>	<i>4</i>	<i>8</i>
<i>X2</i>	<i>2</i>	<i>3</i>
<i>X3</i>	<i>7</i>	<i>2</i>
<i>X4</i>	<i>3</i>	<i>7</i>
<i>X5</i>	<i>9</i>	<i>6</i>
<i>X6</i>	<i>6</i>	<i>9</i>

$$Z = \frac{(X - \mu)}{\sigma}$$

- X = data asli.
- Z = data normalisasi.
- μ = rata rata pada fitur tersebut.
- σ = simpangan baku (standart deviation) fitur tersebut.

Fitur 1

$$\text{Mean } \mu_1 (\text{rata - rata}) = \frac{4 + 2 + 7 + 3 + 9 + 6}{6} = \frac{31}{6} = 5.166$$

$$\text{Simpangan Baku } \sigma_1 = (4 - 5.166)^2 + (2 - 5.166)^2 + (7 - 5.166)^2 + (3 - 5.166)^2 + (9 - 5.166)^2 + (6 - 5.166)^2 = 34.833$$

$$= \sqrt{\frac{34.833}{6}} = \sqrt{5.806} = 2.410$$

Fitur 2

$$\text{Mean } \mu_2 (\text{rata - rata}) = \frac{8 + 3 + 2 + 7 + 6 + 9}{6} = \frac{35}{6} = 5.833$$

$$\text{Simpangan Baku } \sigma_2 = (8 - 5.833)^2 + (3 - 5.833)^2 + (2 - 5.833)^2 + (7 - 5.833)^2 + (6 - 5.833)^2 + (9 - 5.833)^2 = 38.833$$

$$= \sqrt{\frac{38.833}{6}} = \sqrt{6.472} = 2.544$$

Nama	Fitur 1	Fitur 2
X1	$\frac{4 - 5.166}{2.410} = -0.484$	$\frac{8 - 5.833}{2.544} = 0.852$
X2	$\frac{2 - 5.166}{2.410} = -1.314$	$\frac{3 - 5.833}{2.544} = -1,114$
X3	$\frac{7 - 5.166}{2.410} = 0.761$	$\frac{2 - 5.833}{2.544} = -1.507$
X4	$\frac{3 - 5.166}{2.410} = -0.899$	$\frac{7 - 5.833}{2.544} = 0.459$
X5	$\frac{9 - 5.166}{2.410} = 1.591$	$\frac{6 - 5.833}{2.544} = 0.066$
X6	$\frac{6 - 5.166}{2.410} = 0.346$	$\frac{9 - 5.833}{2.544} = 1.245$